CLEAN ENERGY IN INDIGENOUS, RURAL AND REMOTE COMMUNITIES

FUNDING APPLICATION

Instructions

1. This application is for federal funding for activities and projects related to reducing diesel reliance for heat and power, including capacity building, renewable energy and energy efficiency. Your application will be shared with Natural Resources Canada (NRCan), Crown-Indigenous and Northern Affairs Canada (CIRNAC), Indigenous Services Canada (ISC), Environment and Climate Change Canada (ECCC) and Infrastructure and Communities Canada (INFC).

2. Program staff are available to assist you in the application development and submission process. If you have any questions or require assistance, please contact the program at remoteenergy-energieadistance@nrcan-rncan.gc.ca or you can follow up with your existing point of contact within the program.

3. If you have any project documentation that addresses any or all of the Application Form question(s), you may attach and reference these documents instead of filling out these sections. This can include a recently submitted application form or proposal to another funding program, feasibility studies, engagement plans, project governance descriptions, work scope descriptions, project schedules, project budgets, etc.

4. The level of detail in the application should be proportional to the size and complexity of the project. Please indicate if a section does not apply to your project or if you do not have a response to a section. For example, indicate "*not applicable*" or "*This is a project idea and has not been developed further".* There will be opportunities to further provide information throughout the development and assessment process.

5. If any part of this Application contains confidential or proprietary information, please write it in red font.

CLEAN ENERGY IN INDIGENOUS, RURAL AND REMOTE COMMUNITIES

FUNDING APPLICATION

1. Applicant Information		
Applicant Organization	Energy And More Consulting	
Organization Type	Indigenous for-profit organization	
Mailing Address	6 Aishihik Road, Whitehorse, YT, Y1A 3R7	
Business Number *(if applicable)*	*N/A*	
Primary Contact	*Grant Sullivan*	*President*
	Grant_sullivan@hotmail.com	*867-678-5732*
Secondary Contact		*Phone*
Is this the first time your organization has applied for federal clean energy funding?		☒ Yes ☐ No ☐ Unsure If no, please specify program(s)

2. Project Information	
Project Title	**Remote Community Energy & Building Infrastructure Planning AI Model**
Project Location	*Whitehorse, Yukon*
Project Timeline	(2025-04-01) to (2027-03-31)

Total Project Cost	$600,000	**Total Funding Requested**	$600,000

Project Province/Territory *(select all that apply)*

☐BC ☐AB ☐SK ☐MB ☐ON ☐QC ☐NB ☐NS ☐PEI ☐NL ☒NT ☒YT ☒NU

Please specify region, sub-region and/or nation territory ___

Project Description

Provide a full description of the project and key activities. Indicate the work that has been completed, is currently underway, and planned activities to advance the project. Additionally, include the following information based on type of project:

If this is a capacity-building project, please identify how it will help reduce diesel or fossil fuel reliance for power and/or heat and advance community-led clean energy opportunities.

If this is a clean energy project, please identify (if known):

- *the technology being proposed (e.g. wind, solar, hydro, biomass, energy storage, energy efficiency, combinations, etc.);*
- *how it will help improve energy usage within the community and reduce reliance on fossil fuels for heat and power;*
- *the proposed system size including anticipated heat load and/or electricity demand;*
- *whether the project has revenue generating potential;*
- *the proposed plan for managing operation and maintenance costs and maintaining the renewable energy asset throughout its life cycle; and,*
- *whether this is a novel application of the technology, and whether it has been done elsewhere in a similar climate or remote site.*

If this is a research and development project, please identify (if applicable):

- *the generation technology being studied (e.g. small wind, marine, municipal waste conversion, combinations, etc.) and its deployment readiness for Canadian remote communities (e.g. theoretical, prototype, experimental, demonstration, etc.)*
- *scope of the study (e.g. community or industrial scale, estimating heating or electricity loads, energy storage, microgrid reliability, development of a test facility, etc.)*
- *non-technology project considerations (e.g. health or climate implications from diesel, barriers to uptake of renewables, use of traditional knowledge, cultural development models, etc.)*

Project Overview
The Remote Community Energy and Building Infrastructure Planning AI Model Project (the "Project") seeks to address the challenges faced by remote communities in Canada transitioning to net-zero by leveraging AI to create a self-sustaining tool for energy and infrastructure planning while advancing community-led clean energy initiatives. Northern remote communities across Canada are struggling to transition to a net-zero way of life while southern regions develop comprehensive strategies for achieving this goal. Remote communities are not only facing the challenges of transitioning to a net-zero future, but also the direct impacts of climate change such as the impacts of extreme weather on energy access. Limited capacity often requires these communities to rely on external contractors to develop energy of infrastructure plans, which can be expensive and time consuming given the communities' remoteness.

The development of an AI-powered application to help remote communities create their own energy and infrastructure plans would reduce their reliance on outside experts and provide solutions specifically tailored to each community that uses the application. The Project seeks to not only address the immediate energy planning needs of remote communities, but also aims to create a scalable, replicable model that could be adapted and deployed across other remote communities in Canada, potentially amplifying the impact of the tool nationwide.

Project Scope and Components
Task 1: Planning and Research
Task 1 of the Project will lay the groundwork by conducting in-depth research and strategic planning. This will involve an evaluation of existing AI programs to determine how they may be adapted to develop community energy plans at the remote Canadian community level. Feasibility studies and cost-benefit analyses will be conducted to review the development of a standalone AI application. Task 1 will also involve the development of an EOI to gain an understanding of costs related to AI application development.

The development of Community Energy Profiles (CEP) for select communities will take place, to act as inputs for the AI application. A demonstration version of the application will also be developed, to assist in engagement and showcase the technology. Community consultations and stakeholder will be integral during this phase to ensure the AI application is aligned with the community's values, priorities, and unique needs.

Task 2: Development and Testing
Task 2 of the Project will center on the technical development of the full-scale AI application and related testing. During Task 2, AI developers, data scientists, and energy consultants will be engaged to assist in the development of the application and populate it with the necessary data and resources. During Task 2, a comprehensive library consisting of various energy, housing, and community planning resources will be built for the AI application to draw from. These documents may include but are not limited to:

- The Arctic Energy Toolkit;
- The Arctic Remote Energy Networks Academy Program;
- Northern community energy plans;
- Federal funding strategies, reporting, and policies;
- Ministerial mandate letters;
- Territorial energy strategies;
- Electricity and energy strategies;
- General Rate Applications, Power Purchase Agreements; and
- Relevant reports from think tanks and universities;

Once developed, the full-scale application will undergo initial pilot testing in select communities to assess its adaptability to various environments, climates, and community sizes. Once the pilot is complete, feedback will be incorporated into the application to ensure it is ready for real-world use.

Phase 3: Launch and Community Engagement
The final phase of the Project will involve its deployment of the application in target communities, alongside

ongoing support to ensure its successful adoption. The launch will be accompanied by various training and supporting materials to help guide the communities in using the tool effectively. In addition to training materials, the launch will include the establishment of local energy champions who can assist in continued education and serve as in-community facilitators to ensure ongoing success. Post-launch, a system for ongoing support, updates, and troubleshooting will be established to ensure the application remains relevant and functional as communities' needs evolve.

Project Benefits

Provide a description of the project overall result(s) and expected outcome(s). Include details like:

How the project is addressing community, national and/or regional electricity/renewable energy challenges.

- *You may wish to include how the project responds to community needs and/or fits into a community vision/plan.*
- *Examples of challenges: a lack of tools or training for community action, frequent blackouts, end of equipment life, high costs of heat and power, community accessibility.*

Describe the overall benefits and expected results of the project, including who will benefit and how. Please include the expected environmental, social and economic benefits of the project. What would be considered a successful project by the community?

- *Examples: estimated reduction of diesel fuel (litres/year), estimated reduction in greenhouse gas emission (tonnes CO2/year), health benefits (e.g. improved air quality), number and types of jobs created and improved human resource capacity, community energy independence and resiliency, improved reliability of electricity and heat, economic/ financial benefits (e.g., operating savings, lower energy costs).*
- *Describe how the project will contribute to increased capacity within the community and/or region, such as increased institutional capacity and community involvement, improved energy literacy, knowledge and skills development, supporting energy champions, etc.*

If relevant, please describe any past or proposed work related to the project that advances equity, diversity and inclusion in electricity/renewable energy in Canada.

The Project will address critical challenges in remote Canadian communities as they strive toward achieving energy sovereignty and sustainability. These communities often lack the tools and resources to effectively transition to clean energy and face issues such as high energy costs, outdated infrastructure, and limited local expertise. The Project seeks to empower these communities using the AI Application that supports energy and infrastructure planning, while fostering a self-sustaining and community-led approach to the energy transition.

The Project responds to several challenges faced by remote communities, including:

- **Energy sovereignty**: by enabling communities to develop their own energy plans, the Project fosters energy independence by reducing reliance on external contractors.
- **Energy choice**: the Application will give communities greater control over the types of energy sources they wish to explore and incorporate in their own unique systems.
- **Real-time updates**: the Application will function as a dynamic, evolving tool that is reflective of real-time inputs such as changing energy needs, local resources, local priorities, and updated policies.
- **Time and cost savings**: by reducing the reliance on external contractors, the community will save on both time and money that would otherwise be spent on lengthy and expensive external consultation processes.

The Project is expected to result in several benefits, including:

- **Environmental benefits**: the adoption of energy-efficient systems and renewable energy solutions is expected to reduce the reliance on diesel and other fossil fuels in northern remote communities.
- **Economic benefits**: the Project expects to reduce energy costs for remote communities, which face high costs due to various transportation and infrastructure challenges. The Project will improve human resource capacity by developing new skills in energy management, AI tools, and infrastructure planning. Long-term, communities are expected to see savings on energy expenditures.
- **Social benefits**: the Project will strengthen community ownership and experience with energy transition initiatives. It will give communities the tools and agency to direct their own energy futures, fostering a sense of empowerment and resilience. Increased energy literacy and skills development will allow for community members to become energy champions, improving local, territorial, and national leadership.

Project Governance and Team

Provide a summary of the ownership and governance structure of the project. If applicable, include details of Indigenous ownership/partnership of the project and their role in decision-making processes.

Energy and More Consulting, is a northern, wholly Indigenous owned and operated company established under

the laws of Canada.

Grant Sullivan, President of Energy and More Consulting, has expertise and experience in project management, finance, and community liaison. He has a Bachelor of Management Degree (Finance) and has lived in the Beaufort Delta for the majority of his life. He has been active in several Gwich'in organizations as a Gwich'in beneficiary where he gained extensive experience and knowledge of the business community, arctic logistics, and Indigenous governments in the Beaufort Delta region. Grant was a 20/20 Catalyst (2016) and is currently participating as an Energy Champion in the Indigenous Off-grid Diesel Initiative (IODI).

Grant has served as the overall Project Lead/Coordinator for the development and operation of numerous clean energy alternatives and energy efficiency projects across the Beaufort Delta region, as well as other areas in Canada. Grant's experience has involved work on the following projects:

- Inuvik Battery Energy Storage System (BESS) and Birch Wind Project (2022 to Present).
- Inuvik 1 MW Solar Farm (2021 to Present);
- Inuvik High Penetration Renewables Study (2020 to 2021);
- Inuvik Biomass Boiler Installations (2020 to 2022); and
- Biomass Supply Chain Optimization Study (2019);
- Inuvik Satellite Station Facility Solar PV Installation (2018 to 2021);
- Inuvik Commercial Solar PV Installation at Mackenzie Hotel (April 2018 to 2020);
- Inuvik Commercial Solar PV Installation at North Mart (April 2018 to November 2019);
- Inuvik Multi-Residential Solar Net Metering Demonstration (January 2018 to 2020);
- Inuvik High Point Wind Study [for GNWT] (2016 to 2018).

Grant has extensive experience working on studies relating to the development and integration of renewable energy projects in both the Beaufort Delta region and in remote northern communities. Notable studies and related work include:

- Diverging from Diesel for GCI (2016)
- Fossil Fuel Thermal Power Generation Cost in Northern Communities for GCI (2017)
- Arctic Remote Energy Networks Academy (co-lead) for Arctic Council (2016 to 2018)
- Arctic Sustainable Energy Futures Toolkit for Arctic Council (2017 to 2019)
- Beaufort Delta Community Energy Profiles (2017 to 2020)

More information on Grant Sullivan is provided below.

Energy and More Consulting is focused on studying clean energy alternatives and energy efficiency projects in the Beaufort Delta region of Northwest Territories (NWT) as well as in other areas of northern Canada. Energy and More Consulting is focused on working with Indigenous and other northern stakeholders to create sustainable, environmentally responsible and meaningful economic opportunities across the north.

Energy and More Consulting will be responsible for management and coordination of the Project Team and providing oversite for research, engagement, and coordination activities. Energy and More Consulting will be supported by InterGroup Consultants (Project Management Support, Business Case Development) and the University of Waterloo (Technical support related to technical and engineering assessments).

For further details regarding Energy and More Consulting recent project experience see summary for Grant Sullivan, Indigenous Off-Diesel Initiative Champion, provided below. A firm profile and CV can be provided on request.

Provide a summary of the project team. E.g. Who is leading the project? Who will be hired to complete the work (project manager, local project champion, project adviser, consulting engineer, contractors, technology providers, etc.)?

Overall project work will be directed by Energy and More Consulting (Grant Sullivan, Indigenous Off-Diesel

Initiative Champion). Grant will be supported by InterGroup (project management support) and the Future Cities Institute at the University of Waterloo (technical/ engineering assessments).

Key members of the project team have experience working together over a number of projects since 2018/19, including the Inuvik BESS/Birch Wind Project; planning for and installation of solar projects in Inuvik (three commercial developments and 29 residential developments); and completion of a wind feasibility design basis study for the High Point Wind Project.

InterGroup
InterGroup Consultants Ltd (InterGroup) specializes in multi-discipline team management dealing with resource planning and development, project licensing and rate regulation, with a focus on the following activities:

- Coordinating and managing environmental assessment studies and providing expert technical services related to socio-economic impact assessment and public engagement, and also facilitating coordination across multiple technical discipline experts, as required.
- Assisting clients with the planning and development of major resource projects, including providing project management and coordination, planning and economic feasibility studies, and negotiation and facilitation
- Providing expert advice and support related to the development and review of rates for regulated utilities. This includes extensive and ongoing experience providing economic and financial and other advice to Crown utilities in Yukon (working with Yukon Energy Corporation since 1987), Northwest Territories (working with Northwest Territories Power Corporation since 2000) and Nunavut (working with Qulliq Energy Corporation since 2009).

InterGroup's team has acquired extensive knowledge regarding fuel costs related to generating electricity and heating in northern communities, including communities connected to predominantly hydro-electric grid power, as well as isolated or off-grid communities that rely on diesel or natural gas generation. InterGroup has also provided ongoing senior advice, technical and project management support with Grant Sullivan's assessments for the residential solar net metering development in Inuvik, NT; the development of two commercial solar projects in Inuvik; the 60 kW ISSF site solar development near Inuvik; the 150 kW solar development IPP in Aklavik; and the 1 MW solar development IPP in Inuvik. InterGroup also provided business case and project management support for the Inuvik High Point Wind Project Design Basis and Business Case study; the Inuvik High Penetration Renewables Study; and the Inuvik BESS/Birch Wind Project.

InterGroup will provide Energy and More Consulting with support and strategic guidance in the following areas:

- Project management support;
- Support for negotiation of agreements;
- Business case development/ review of project economics and strategic planning/ approach.

Future Cities Institute, University of Waterloo The University of Waterloo's Future Cities Institute (FCI) collaborates with a diverse range of partners, from large multinational corporations and municipalities to smaller Indigenous communities and hamlets, focusing on critical areas such as housing, mobility, sustainable infrastructure, and the integration of artificial intelligence within municipal and hamlet settings. Through operational, applied research, FCI leverages its expertise to address complex, real-world challenges defined by city or community representatives and other partners. Current projects, in collaboration with municipalities and community partners, showcase a commitment to impactful research and development, enhancing sustainable growth and resilience through practical, innovative solutions. This approach not only drives forward-thinking advancements but also ensures that researchers, students and partners work together to co-create solutions that are meaningful and effective within the unique context of each community.

The FCI will support this project by integrating resources from a global network of interdisciplinary researchers, industry leaders, students, and community members focused on sustainable municipal development.

More specifically FCI will provide access to UWaterloo's advanced computing resources with AI/ML

capabilities, specialized software for AI and language model development, and powerful data analysis tools. Ranked as Canada's #1 comprehensive research university (Research Information 2023), UWaterloo also brings expertise from Canada Excellence Research Chairs in Human-Centred Robotics, Machine Intelligence, Developmental Science, and Data Intelligence to advance project goals.

In addition, the FCI will develop a team of world-class researchers, post-doctoral scholars, graduate students and undergraduate students to lead the technical and engineering components of this important project. Members of this team may include (but are not limited to) representatives from the following faculties and institutes across campus:

- **Waterloo Data and Artificial Intelligence Institute (Waterloo.AI):** Waterloo.AI offers participants intensive, short courses (3-4 days) led by top instructors in AI, allowing project team members to deepen their technical expertise. These courses are tailored to bridge academic AI knowledge with practical applications, providing participants the skills to integrate new AI techniques directly into industry contexts.
- **UWaterloo Faculty of Engineering:** As the largest engineering faculty in Canada, UWaterloo's Faculty of Engineering supports the project by providing resources, expertise, and tools required for AI-driven urban solutions. The faculty fosters opportunities for students to engage with AI through applied design projects, educational workshops, and hackathons, directly contributing to innovative project outcomes.
- **UWaterloo Faculty of Mathematics:** The Faculty of Mathematics, home to the largest Computer Science department in Canada, offers additional resources and support for complex computational projects. With advanced capabilities in data science and AI, the faculty is integral in applying rigorous data intelligence approaches to urban challenges.

In tables below, describe key team members involved in the project (e.g., their experience, role and expertise they will bring). If applicable, describe any legal background and who will retain any intellectual property for the project.

Provide sufficient detail for the program to assess whether the team has the necessary capability and capacity (e.g., management, technical, etc.) for the proposed work, but do not include resumés.

Add/delete rows as needed.

Team Member		Organization	Role	Phone Number	Email
1	*Grant Sullivan*	*Energy and More Consulting Energy Corporation*	*President, Indigenous Off-Diesel Initiative Champion*	(867)678-5732	grant_sullivan@hotmail.com

Grant Sullivan (President, Energy and More Consulting) will be the Project Manager, and responsible for management and coordination of the Project Team and providing oversite for design, procurement, permitting, installation and commissioning activities, as well as ongoing maintenance and reporting activities. Key areas of responsibility include:

- Main point of contact providing direction to Project Team Members & overall team coordination.
- Lead discussions with relevant parties related to negotiation of agreements; determining contract approach/ framework; and contract/ agreement finalization.
- Lead interactions with federal and other funding partners; including reporting as required.

Mr. Sullivan has expertise and experience in project management, finance and community liaison. He has a Bachelor of Management Degree (Finance), and has lived in the Beaufort Delta region for the majority of his life. He has been active in several Gwich'in organizations as a Gwich'in beneficiary where he gained extensive experience and knowledge of the business community, arctic logistics and Indigenous governments in the Beaufort Delta region.

Mr. Sullivan has served as overall Project Lead/Coordinator for a number of renewal energy projects advanced by either Nihtat Corporation or Nihtat Energy Ltd (NEL) since 2018 (see below for summary focused on Beaufort Delta projects). Work on each of these projects was initially advanced by Nihtat Corporation and then undertaken by NEL starting in May 2019. Key relevant projects are noted below.

- Inuvik High Point Wind Study [for GNWT] (2016-2018): Feasibility study for the Government of Northwest Territories (August 2016 to April 2017). Feasibility work for GNWT towards development of a Business Case and Design Basis for the Inuvik Wind Project. Mr. Sullivan provided oversight for work related to the geotechnical survey in Inuvik, site preparation and installation of a met tower at the proposed project site, environmental baseline data acquisition and review, two GNWT Workshops and two community engagement sessions, followed by completion of the design basis and presentation of the business case.

- Inuvik High Penetration Renewables Study (2020-2021): Pre-feasibility assessment for potential addition to the Inuvik grid of up to 1-3 MW of wind generation; up to 1 MW of solar; and energy storage (the "Inuvik Grid High Penetration Renewable Pre-feasibility Study").

- Inuvik BESS/Birch Wind Project (2022 to Present). Feasibility assessment of an 8-11 MWh battery/ microgrid and the 4.2 MW Birch Wind Project, and related opportunity for a renewable hydrogen development project.

- Commercial Solar Developments in Inuvik: Planning, design, procurement and installation of 250 kW DC of ground mounted solar PV at the North Mart Store (2019); 144 kW DC of ground mounted solar PV at the Mackenzie Hotel (2020); 60 kV of ground mounted solar PV at the ISSF facility near Inuvik (2021); and 1 MW Solar Farm IPP (2021 to Present) for sale of electricity to Northwest Territories Power Corporation through a Power Purchase Agreement.

- 150 kV Solar IPP in Aklavik: design, procurement and installation of up to 265 kW DC (restricted to 150 kw AC) of solar PV capacity to the Aklavik community grid in 2020-2021, for sale of electricity to Northwest Territories Power Corporation through a Power Purchase Agreement (completed in September 2020). The project has been constructed and is in the process of being commissioned (expected in service in 2024).

Grant has extensive experience working on studies relating to the development and integration of renewable energy projects in both the Beaufort Delta region and in remote northern communities. Notable studies and related work include:

- Diverging from Diesel for GCI, as Executive Director of GCI (2016)
- Fossil Fuel Thermal Power Generation Cost in Northern Communities for GCI, as Executive Director of GCI (2017)
- Arctic Remote Energy Networks Academy (co-lead) for Arctic Council, as Executive Director of GCI (2016 to 2018)
- Arctic Sustainable Energy Futures Toolkit for Arctic Council, as Executive Director of GCI (2017 to 2019)
- Beaufort Delta Community Energy Profiles for NEL, as President of NEL (2017 to 2020)

A firm profile and CV can be provided on request.

2	*Cam Osler*	*InterGroup Consultants Ltd*	*Project Management Support, Business Case Development*	*204-942-0654*	Cam@intergroup.ca

Mr. Osler will provide senior project management support and lead business case development.

He has over 50 years of experience in regulatory economics, including revenue requirement, cost of service, and rate design providing case management support during regulatory review processes and providing expert

testimony before regulatory tribunals. Other experience includes team leadership on multidisciplinary management teams, socio-economic and environmental assessment and related public consultation, strategic planning and assessment, and participation in negotiation of compensation agreements for First Nations. He has provided ongoing senior advice, technical and project management support regarding Mr. Sullivan's assessments for the residential solar net metering development in Inuvik, NT; the development of two commercial solar projects in Inuvik; the 60 kW ISSF site solar development near Inuvik; and the 150 kW solar development in Aklavik. He also provided business case and project management support for the Inuvik High Point Wind Project Design Basis and Business Case study; the Inuvik High Penetration Renewables Study; and the Inuvik BESS/Birch Wind Project.					
3	*Mona Pollitt-Smith*	*InterGroup Consultants Ltd*	*Project Management Support, Business Case Development*	*204-942-0654*	mona@intergroup.ca
Ms. Pollitt-Smith will provide project management support and support business case development. Mona has over fifteen years of experience with InterGroup working on regulatory, resource planning and development projects, and environmental assessment and licencing reviews; and has experience coordinating and leading preparation of a variety of complex regulatory filings and providing support throughout the regulatory review process, as well as experience supporting negotiation of power purchase agreements and project agreements with First Nations. She has provided ongoing advice, technical and project management support regarding Mr. Sullivan's assessments for the residential solar net metering development in Inuvik, NT; the development of two commercial solar projects in Inuvik; the 60 kW ISSF site solar development near Inuvik; and the 150 kW solar development in Aklavik; as well as project management support for the Inuvik High Point Wind Project Design Basis and Business Case study; the Inuvik High Penetration Renewables Study; and the Inuvik BESS/ Birch Wind Project.					
4	*Eryn Stewart*	*Future Cities Institute, University of Waterloo*	Technical Lead	*(905) 531-4114*	*eryn.stewart@uwaterloo.ca*
Eryn Stewart serves as the Managing Director of the Future Cities Institute (FCI) at the University of Waterloo where she provides overall strategic leadership, management and administration for all FCI programs and services across the University of Waterloo. She has been instrumental in the development of FCI's growing member network which includes academic collaborators from across UWaterloo's six faculties. Before joining the FCI, Eryn was the co-founder and Managing Director of Indigenous Clean Energy (ICE), a community-driven enterprise advancing Indigenous inclusion in Canada's clean energy future. At ICE, she played a transformative role in fostering partnerships and empowering Indigenous communities to lead sustainable energy projects across the country. In 2017, Eryn was one of twenty emerging leaders selected by the Arctic Council to participate in the Arctic Remote Energy Network Academy – a program to promote information sharing across the Arctic on the integration of renewable energy in remote communities. Recognized as a Top 30 Under 30 Sustainability Leader and a recipient of the University of Waterloo Young Alumni Inspiration Award, Eryn brings a proven track record of driving impactful change to her leadership at FCI, along with expertise in community energy planning, community engagement, and energy efficiency in northern and remote communities.					
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3. Project Engagement
Indigenous and Community Engagement
Describe which communities will participate in the project, the level of community involvement both during the project and beyond, the ongoing relationship between you as the applicant organization with the communities, and completed and/or planned community engagement efforts. Indicate the level of support from the community(ies) and alignment with community plans and values. Include any confirmed written support, funding, leadership, etc. *What opportunities will be made to allow the views of under-represented groups (e.g. women, gender-diverse people, youth, Elders, 2SLGBTQ+, people with disabilities, etc) to be shared and incorporated into the project?* The proposed project will actively engage with key Indigenous communities and local governments, including the City of Iqaluit, the Council of the Haida Nation, and the Town of Inuvik. These communities will play a vital role in shaping the project's development by providing input, sharing local knowledge, and ensuring that outcomes align with community priorities. Engagement efforts will focus on reviewing the AI model as it progresses, incorporating relevant grid data, and ensuring the integration of existing energy plans with community-specific needs. The collaborative review process will also serve as a foundation for planning implementation activities tailored to each community's unique energy landscape. **Commitment to Meaningful and Ongoing Engagement** Energy and More Consulting prioritizes fostering strong, respectful, and mutually beneficial relationships with Indigenous governments, community organizations, and residents. Engagement will go beyond information-sharing by ensuring communities have an active voice in decision-making. Our approach includes: • **Early and Continuous Engagement**: Community input will be sought at key stages of the project to ensure alignment with local priorities, traditional knowledge, and long-term sustainability goals. • **Transparent Communication**: Regular updates, workshops, and community meetings will provide opportunities for meaningful dialogue and collaboration. • **Local Capacity Building**: Training initiatives, workshops, and knowledge-sharing sessions will be developed to enhance community energy literacy and build local expertise in clean energy solutions. • **Economic Opportunities**: The project will prioritize Indigenous-led businesses and contractors to maximize local employment and economic benefits. **Community Support and Alignment with Local Plans** *T*he project aligns with local energy strategies, sustainability goals, and community visions for energy independence. Support letters from Indigenous governments, local leadership, and

relevant stakeholders will be sought to demonstrate community buy-in. Engagement efforts will also leverage existing partnerships and explore new collaborations to strengthen community participation.

Anticipated Partners and Supporters

Please list the names of any relevant partner organizations or individuals and indicate their role in the project. This would include potential funders, governments, utilities, etc.

For projects involving the demonstration or deployment of a clean energy technology, describe the partnerships with Provincial/Territorial Government and Utility, who has been engaged and their level of support.

Example: are there memorandums of understanding (MOUs), joint ventures (JVs) or power/energy purchase agreements under development; is the project outlined in a community energy plan; are environmental assessments/permits underway; if no engagement has occurred is there an engagement plan in place?

Partner/Supporter	Description of Role
City of Iqaluit	Attached letter of support
Haida Nation	Attached letter of support
Town of Inuvik	Attached letter of support

4. Project Workplan

Workplan Details

Please attach any detailed project plan documents (e.g. project timeline, project budget). You may reference these documents instead of filling out this section.

Using the table below, describe how the project will be carried out including; a list of project tasks; a high-level description for each; who is expected to complete them; what are the anticipated outputs or results (e.g. milestones); what is the timeline for each task; and the estimated cost.

If applicable, please include details on the required permits, environmental assessments and/or regulatory approvals for construction and operation of the project.

Add/delete rows as needed.

The Workplan below reflects the three identified tasks with each Task's related deliverables and timeline; within each task, responsibilities and estimated costs are described separately for Energy and More Consulting, InterGroup and University of Waterloo.

Project Task	Description	Team Member or Organization Responsible	Output/result	Start and Completion Date	Estimated Cost ($)
Task 1: Planning and Research		Energy and More Consulting			**$50,000**
Review of existing AI applications	Research and evaluate available AI solutions that could be tailored to energy and infrastructure planning.	Energy and More Consulting			$10,000
Review Intellectual property governance issues	Review University of Waterloo Intellectual property guidelines	Energy and More Consulting			$20,000
Develop Community Energy Plans and Related Community engagement	Development of Community Energy Plans and initial community engagement to gather inputs into application	Energy and More Consulting			$20,000
Task 2: Development and Testing					**$340,000**
Contract services for application	Engagement of AI developers, data scientists, and energy consultants to help develop the application.	University of Waterloo			$190,000
Build library and populate application	Compilation of energy planning resources, strategies, and policy guidelines and integration into the application.	University of Waterloo			$50,000
Testing & revisions	Conduct pilot testing in select communities and incorporation of feedback into application.	University of Waterloo			$100,000
Task 3: Launch and Community Engagement					**$210,000**

Launch application	Deployment of AI application in communities, accompanied by training and support materials.	Energy and More Consulting			$30,000
Community engagement	Engagement associated with various communities to develop communities' relationships with the application and showcase how it can be used.	Energy and More Consulting			$60,000
Project Management & Reporting	Associated project management (salary dollars for Grant Sullivan over 2 years) and reporting.	Energy and More Consulting and InterGroup			$120,000
Total All Tasks					**$600,000**

5. Project Risks and Mitigation

Using the table below, please describe any risks associated with the project (e.g., financial, scope, timeline, technical, consultation, market, etc.) and include the estimated likelihood (high, medium, low) and impact of the risk, as well as the mitigation measure that will be taken.

Description of Risk	Likelihood and Project Impact	Mitigation Measure
Technical Feasibility of AI Model: The AI model may face challenges in integrating diverse energy and infrastructure data sources across different remote communities, leading to limitations in its effectiveness.	**Medium Likelihood, High Impact**: AI-based energy planning tools require reliable data inputs, and variations in data availability and quality across different communities may impact performance.	A phased development approach will be used, starting with pilot testing in select communities. Continuous refinement through feedback loops will ensure the model adapts to different data inputs and improves its predictive accuracy. Collaborations with the university and AI experts will enhance model robustness.
Community Adoption and Engagement: Communities may be hesitant to adopt the	**Medium Likelihood, High Impact**: Limited digital literacy and unfamiliarity	The project will include extensive community consultations, user-friendly training materials, and

AI-based planning model due to concerns about digital literacy, local energy priorities, and skepticism about AI-driven decision-making.	with AI tools may slow adoption and reduce engagement from Indigenous and remote communities.	dedicated capacity-building initiatives. Local energy champions will be identified to provide in-community support and promote adoption. Training workshops will be held to ensure ease of use.
Data Security and Privacy Concerns: The AI model requires sensitive community energy data, raising concerns about data privacy, security, and ownership.	**Low Likelihood, High Impact**: If data governance is not properly managed, communities may be reluctant to share critical energy information, reducing the model's effectiveness.	Data protection policies and governance frameworks will be implemented, ensuring communities retain ownership and control over their data. Secure cloud storage, encryption protocols, and compliance with federal privacy regulations will be prioritized.
Project Funding and Long-Term Sustainability: Securing full funding and ensuring long-term operational sustainability of the AI model beyond the project's initial implementation phase.	**Medium Likelihood, Medium Impact**: Without sustained funding, future updates, maintenance, and expansion of the AI model to additional communities may be at risk.	The project team will actively pursue diverse funding sources, including private sector partnerships, government grants, and Indigenous-led investment opportunities. A revenue model will be explored for long-term sustainability, including potential licensing to other communities.

Description of Risk	Likelihood and project impact	Mitigation measure
Describe the risk to the successful execution of the project	*Describe the estimated likelihood of the risk occurring and the impact to the project should it occur*	*Describe the mitigation measure taken and estimate the residual risk to project (low, medium, or high)*

Residual Risk Level	
Low	The residual risk has little potential for impact on project success. The Applicant is confident the mitigation measures in place effectively manage the risk.
Medium	The residual risk has some potential for impact on the project. With mitigation measures in place, the potential impact is not critical to project success or is very unlikely to occur.
High	The residual risk has significant potential to impact project success with outlined mitigation measures in place.

6. Additional Information (optional)
☒ Yes ☐ No Supporting documents have been attached to this Application.
List file names of attached documents, for example, permits, land access approvals, environmental assessments, support letters, studies, etc. • Attachment 1 – Relevant Inuvik Experience Summary

7. Project Financials:

Please provide sources of funding of the proposed project. Leave blank any sections that do not apply to you. Insert additional rows as needed. Organize by fiscal years (i.e. 2022-23 is April 1, 2022 to March 31, 2023). Please note there will be an additional Budget Template to complete at a later stage to indicate cost categories of the proposed project.

	2022-2023	2023-2024	2024-2025	2025-2026	2026-2027	Total
Contributions						
Program funding sought:			500,000	100,000		600,000
Other Government of Canada Funding: • Impact Canada – Indigenous Off-Diesel Initiative Please indicate the status of funding (applied, pending, confirmed).						
Other Provincial, Municipal, Territorial or Indigenous government funding:						
Applicant Organization						
Funding from other sources: • Please identify source						
In-Kind: • List organization(s)						
Total contributions per year:			500,000	100,000		600,000

8. Applicant Attestation

By submitting this project application, the Applicant:

1. Attests that they have read and agree to the terms and conditions in the Applicant Guide.
2. Attests that it is legally registered or incorporated in Canada (does not apply to governments).
3. Attests that the information provided is true and accurate to the best of their knowledge.
4. Understands that any costs incurred for the submission of the Application Form are at the Applicant's own risk.
5. Understands that CIRNAC and NRCan reserve the right to alter the currently envisaged process, and deadlines, or to cancel the entire application process at its sole discretion.
6. Understands that project funding decisions will only be made following receipt, review, and selection of projects and the successful completion of due diligence.
7. Understands and acknowledges that no liability, commitment or obligation exists on the part of CIRNAC and NRCan to make a financial contribution to the project until a written funding agreement is signed by both parties.
8. Attests that it is the owner of all information - proprietary, confidential or otherwise - provided as part of the proposal submission, or, if the information belongs to another party, that it has obtained written consent to disclose the information to CIRNAC and NRCan.
9. Understands that federal reviewers are bound by the requirements of the *Access to Information Act* and the *Privacy Act* regarding the treatment of confidential information.

By checking below, the Applicant agrees that the Government of Canada may share this application and any other information provided as supplemental material with other potential funding entities in an effort to better support projects. Please indicate which of the following you consent to having your proposal shared with:

a) Other Government of Canada Departments or Agencies;	☒ **Yes** ☐ **No**
b) Provincial, Territorial and Municipal Governments;	☒ **Yes** ☐ **No**
d) Local and/or Regional Utility; and	☒ **Yes** ☐ **No**
e) Not-for-profit sector or foundations (e.g., Federation of Canadian Municipalities)	☒ **Yes** ☐ **No**

9. Conflict of Interest Attestation

Definition: *A person is in a conflict of interest when they exercise an official power, duty or function that provides an opportunity to further their private interests or those of their relatives or friends or to improperly influence another person's private interests.*

The following questions are to be used as guidelines to help you assess if a real, potential or perceived conflict of interest exists. If "Yes" for any of the below, the Program will contact you to discuss further

(i.e. not an automatic project disqualification). To the best of your knowledge:			
Are any current or former Government of Canada employees, excluding your current point of contact within the program, working or associated with the development of this proposal or project?		No	
Will a current or former public servant or any of their relatives or friends be receiving any personal or financial benefit of any kind as a result of the Government of Canada entering into an agreement with your organization?		No	
Have you, any individual working in your organization, your partners' organizations, or any individual working in their organization, formerly provided consultancy services to the Minister or departments that are related to this project, particularly any services associated with developing the proposal?		No	
Please sign below to confirm having read and understood the statements above			
Signature:		**Date:**	2025-03-04
Name of Duly Authorized Officer:	**Grant Sullivan**		
Title:	**President**		